

# **Exchange rate and Production network**

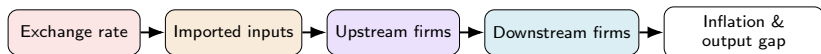
Bonn Macro Internal Seminar

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# Motivation



Firms buy inputs mostly from *other domestic firms*, not just labor — a cost shock to one sector mechanically hits everyone downstream

Imported inputs are the norm (Chile: up to 19.5% in Manufacturing) — so an FX move is, for the firm, a domestic cost shock

Network models get this propagation right but ignore the exchange rate; open-economy models get the exchange rate right but ignore the network

**Does accounting for production networks change the optimal exchange-rate regime?**

# Literature

**Network side (closed economy):** Rubbo (2024); La'O & Tahbaz-Salehi (2022); Pasten, Schoenle & Weber (2020); Ozdagli & Weber (2026); Ghassibe (2021)

**Open-economy side (no network):** Galí & Monacelli (2005); Fanelli & Straub (2021); Schmitt-Grohé & Uribe (2003); Gopinath & Itskhoki (2022); Amiti et al. (2019); Auer, Levchenko & Sauré (2019); Itskhoki & Mukhin (2023); Bianchi & Coulibaly (2025); Coulibaly (2023)

**Open economy with network:**

**Gnolato, Montes-Galdón & Stamato (2025, ECB); Kalemli-Özcan et al. (2025)** — open economy *with* network (tariffs), fixed Taylor rule, no FX-regime choice<sup>1</sup>

**Qiu, Wang, Xu & Zanetti (2026, JME)** — closest paper: derives the open-economy DC index directly extending Rubbo, WIOD 43-country calibration; but *static* (no NFA/persistence), no UIP, and no FX-regime choice

**Gap:** network + open economy + FX regime choice, together.

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<sup>1</sup>Details & what these papers cite for open-economy production networks: Appendix.

# This Paper

Extend Rubbo (2024) → SOE: imported inputs, UIP and debt-elastic premium

**Three literatures, none overlapping fully** (network only; open economy only; network + open economy but no regime choice) — **first to put all three together**

**Calibrate:** Chile's real IO table, also Korea, Czechia

Ask: rank FX regimes? How does the network amplify foreign shocks? How do sector heterogeneity and network structure matter?

**Result: Managed < Float  $\ll$  Peg** (welfare loss  $10.96/25.15/90.45 \times 10^{-4}$ )

**Seven shocks, two kinds — keep this distinction throughout:** **cost-push** (sector TFP  $A_i$ , import price  $PF$ ) enters marginal cost directly and breaks divine coincidence; **demand** (foreign demand  $D^*$ , export price/ToT  $PX$ , risk-premium/UIP  $RP$ ) moves  $\tilde{y}_t$  without a first-order cost-push term of its own — but under a **fixed** exchange rate, the interest-rate response needed to defend  $\hat{s}_t=0$  turns even a pure demand shock into a large output-gap cost (see Peg, throughout)

# Households

Representative household chooses  $\{C_t, L_t, B_t, B_t^*\}$  to maximize lifetime utility, subject to a budget constraint with a domestic bond  $B_t$  and a foreign bond  $B_t^*$ :

$$E_0 \sum_{t=0}^{\infty} \beta^t \left[ \frac{C_t^{1-\gamma}}{1-\gamma} - \frac{L_t^{1+\varphi}}{1+\varphi} \right]$$

subject to

$$P_t^C C_t + B_t + S_t B_t^* = W_t L_t + \Pi_t + (1+i_{t-1})B_{t-1} + (1+i^*)[1-\psi(\hat{b}_{t-1}^* - \bar{b}^*)]S_t B_{t-1}^*$$

Consumption is CES across home/foreign goods, Cobb-Douglas across domestic sectors:

$$C_t = \left[ \omega^{1/\eta} (C_t^H)^{\frac{\eta-1}{\eta}} + (1-\omega)^{1/\eta} (C_t^F)^{\frac{\eta-1}{\eta}} \right]^{\frac{\eta}{\eta-1}}, \quad C_t^H = \prod_i (C_{it}^H)^{\beta_i^H}$$

# Firms: Technology

Firms in sector  $i$  share Cobb-Douglas technology (labor, domestic inputs, imports) and price under monopolistic competition:

$$Y_{ift} = A_{it} L_{ift}^{\alpha_i} \prod_j X_{ijft}^{\Omega_{ij}^H} M_{ift}^{\Omega_i^F}$$
$$MC_{it} = \frac{W_t^{\alpha_i} \prod_j P_{jt}^{\Omega_{ij}^H} (S_t P_t^{F*})^{\Omega_i^F}}{A_{it}}$$

**Two cost-push channels:**  $MC_{it}$  moves with  $A_{it}$  (TFP) and  $S_t$ , the exchange rate (via the import cost  $S_t P_t^{F*}$ ).

The theory holds for any number of sectors  $N$ ; the quantitative results use  $N = 3$  (Resource, Manufacturing, Services), collapsed from Chile's 12-sector national IO table.

# Firms: Price Setting

Calvo pricing through the network delivers the (vector) Phillips curve:

$$\boldsymbol{\pi}_t = \rho(I - \mathcal{V})\mathbb{E}_t\boldsymbol{\pi}_{t+1} + \mathcal{B}(\gamma + \varphi)\tilde{y}_t - \mathcal{V}\boldsymbol{\chi}_t + \Gamma\Delta e_t$$

**Same amplifier, two shocks:**  $\mathcal{B}, \Gamma$  are the *same* rigidity-adjusted operator  $\tilde{A} \equiv \Delta(I - \Omega^H \Delta)^{-1}$ , applied to labor share  $\alpha$  vs. import share  $\omega_F$ :

$$\mathcal{B} = \frac{\tilde{A}\alpha}{\kappa}, \quad \Gamma = \frac{\tilde{A}\omega_F}{\kappa}$$

where:

|                                                |                                       |                                      |                      |
|------------------------------------------------|---------------------------------------|--------------------------------------|----------------------|
| $\boldsymbol{\pi}_t, \tilde{y}_t$ : infl., gap | $\boldsymbol{\chi}_t$ : TFP cost-push | $\Delta e_t$ : exch. rate chg.       | $\kappa$ : GE scalar |
| $\mathcal{B}, \Gamma$ : PC slope, FX pass-thru | $\mathcal{V}$ : TFP cost-push matrix  | $\tilde{A}$ : rigidity-adj. Leontief |                      |

# Firms: Production Network

The **Leontief inverse** sums direct and indirect linkages:

$$(I - \Omega^H)^{-1} = I + \Omega^H + (\Omega^H)^2 + \dots$$

applied to consumption shares  $\beta^H$  (Households) it gives:

$$\lambda_D^\top = (\beta^H)^\top (I - \Omega^H)^{-1}, \quad \mathcal{M} = (I - \Omega^H)^{-1} \omega_F, \quad \mathcal{M}^X = (I - \Omega^H)^{-1} \omega_X$$

where:

$\lambda_{D,i}$ : **Domar weight** — sector  $i$ 's total weight in final demand, network-adjusted (can exceed 1 for downstream sectors)

$\mathcal{M}_i$ : **Import centrality** — sector  $i$ 's total (direct + indirect) exposure to import costs

$\mathcal{M}_i^X$ : **Export centrality** — sector  $i$ 's total (direct + indirect) exposure to export demand

**Why left vs. right multiply:**  $\lambda_D$  asks "how much of *final demand* ultimately flows through sector  $i$ ?" — a question about a row vector of weights pulled *backward* from consumption, hence  $(\beta^H)^\top (\cdot)$  on the left.  $\mathcal{M}, \mathcal{M}^X$  ask "how much of a *cost/demand shock at the border* reaches sector  $i$ ?" — a question about a column vector pushed *forward* through supply chains, hence  $(\cdot) \omega_F$  on the right. Same Leontief inverse, opposite direction of the question

# Foreign Sector

Law of one price and UIP with a debt-elastic NFA stabilizer  $\psi(\cdot)$  and an exogenous risk-premium shock  $RP_t$ :

$$P_t^{FH} = S_t P_t^{F*}, \quad i_t = i^* + \mathbb{E}_t \Delta e_{t+1} + \psi(\hat{b}_t^*) + RP_t$$

NFA accumulates the trade balance:

$$\hat{b}_t^* = \frac{1 + i_{t-1}}{\Pi_t^C} \hat{b}_{t-1}^* + \frac{P_t^H \sum_i EX_{it} - P_t^{FH} IM_t}{P_t^C}, \quad EX_{it} = \kappa_{EX,i} D_t^* PX_t \left( \frac{P_{it}}{P_t^{FH}} \right)^{-\theta^*}$$

**Two channels** to the domestic economy:

$$\hat{s}_t + \hat{p}_t^{F*} \rightarrow MC_{it} \rightarrow \pi_{it} \quad (\text{cost-push}) \quad \hat{s}_t \rightarrow EX_{it} \rightarrow \tilde{y}_t \rightarrow \pi_{it} \quad (\text{demand})$$

where:

$P_t^{F*}$ : foreign price of imports  
 $\psi(\cdot)$ : debt-elastic NFA stabilizer  
 $\hat{s}_t \equiv \log(S_t/\bar{S})$ : log exch. rate

$RP_t$ : risk-premium shock  
 $\hat{b}_t^*$ : NFA (% GDP)  
 $\Delta e_{t+1} \equiv \mathbb{E}_t \hat{s}_{t+1} - \hat{s}_t$ : exp. depreciation

$D_t^*$ : foreign demand shock  
 $\kappa_{EX,i}$ : export scale, sector  $i$   
 $PX_t$ : export price/ToT shock  
 $\theta^*$ : export demand elasticity  
 (hats: log-deviation from steady state, throughout)

# Monetary Policy Regimes

## Free float

$$\hat{i}_t = \phi_\pi \pi_t^{DC} + \phi_y \tilde{y}_t$$

The exchange rate adjusts endogenously.

## Hard peg

$$\hat{s}_t = 0$$

Adjustment occurs through domestic inflation and output.

## Managed float

$$\hat{i}_t = \phi_\pi \pi_t^{DC} + \phi_y \tilde{y}_t + \phi_s \hat{s}_t$$

The central bank partially stabilizes the exchange rate.

# Equilibrium System

Phillips curve:

$$\pi_t = \rho(I - \mathcal{V})\mathbb{E}_t\pi_{t+1} + \mathcal{B}(\gamma + \varphi)\tilde{y}_t - \mathcal{V}\chi_t + \Gamma\Delta e_t$$

IS curve:

$$\tilde{y}_t = \mathbb{E}_t\tilde{y}_{t+1} - \frac{1}{\gamma}(i_t - \mathbb{E}_t\pi_t^C - r_t^{\text{nat}}) - \frac{\eta(1 - \omega)}{\gamma}\mathbb{E}_t\Delta(\hat{s}_t + \hat{p}_t^{F*} - \hat{p}_t^H)$$

Foreign sector:

$$i_t = i^* + \mathbb{E}_t\Delta e_{t+1} + \psi(\hat{b}_t^*) + RP_t$$

Policy rule:

$$\hat{i}_t = \begin{cases} \phi_\pi \pi_t^{DC} + \phi_y \tilde{y}_t & \text{Float} \\ \text{residual } (\hat{s}_t = 0) & \text{Peg} \\ \phi_\pi \pi_t^{DC} + \phi_y \tilde{y}_t + \phi_s \hat{s}_t & \text{Managed} \end{cases}$$

**Not over-identified:** law of one price ( $P_t^{FH} = S_t P_t^{F*}$ , Foreign Sector slide) is a *definition* substituted into  $\chi_t$ /the IS curve's real-exchange-rate term, not an extra equilibrium condition — the system is  $N$  Phillips curves + IS + UIP + policy rule solving for  $N$  sectoral  $\pi_{it}$ ,  $\tilde{y}_t$ ,  $i_t$ ,  $\hat{s}_t$ : equal counts, one equation per unknown.

| Shock | $A_1$ | $A_2$ | $A_3$ | $PF$ | $D^*$ | $PX$ | $RP$ |
|-------|-------|-------|-------|------|-------|------|------|
|-------|-------|-------|-------|------|-------|------|------|

s.d. (all)

1%

# Calibration: Chile Data

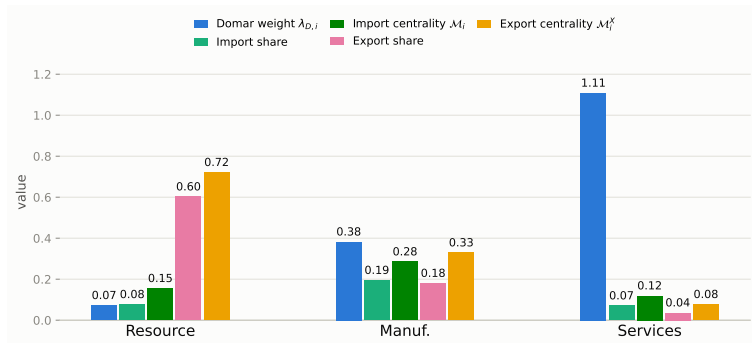
| Param.                               | Description                | Value               | Source                                 |
|--------------------------------------|----------------------------|---------------------|----------------------------------------|
| $\alpha_1, \alpha_2, \alpha_3$       | Labor share                | 0.503, 0.359, 0.604 | Data: BCCh IO                          |
| $\Omega_1^F, \Omega_2^F, \Omega_3^F$ | Import cost share          | 0.077, 0.195, 0.070 |                                        |
| $\beta_1^H, \beta_2^H, \beta_3^H$    | HH consumption share       | 0.027, 0.229, 0.744 |                                        |
| Export share <sub>1,2,3</sub>        | Export share of output     | 0.602, 0.180, 0.036 |                                        |
| $\delta_1, \delta_2, \delta_3$       | Calvo reset prob.          | 0.90, 0.31, 0.16    | Lit.: Dhyne et al. (2005); N-S (2008)  |
| $\psi$                               | Risk-premium elasticity    | 0.020               | Key mechanism param.<br>(see appendix) |
| $\phi_\pi$                           | Taylor: inflation resp.    | 1.50                |                                        |
| $\phi_y$                             | Taylor: output-gap resp.   | 0.50                |                                        |
| $\phi_s$                             | Taylor: FX-smoothing resp. | 0.30                |                                        |

$\Omega^H$  (order: Res., Man., Serv.; source: BCCh IO table):

$$\Omega^H = \begin{pmatrix} 0.075 & 0.153 & 0.193 \\ 0.099 & 0.202 & 0.145 \\ 0.002 & 0.058 & 0.266 \end{pmatrix}$$

**Standard literature values** (not paper-specific; sources in Appendix):  $\beta=0.99$ ,  $\gamma=1.00$ ,  $\varphi=2.00$ ,  $\eta=1.50$ ,  $\varepsilon=8.00$ ,  $\theta^*=2.00$ ,  $\beta^F, \text{tot}=0.10$ .

# Network Properties: Import & Export Exposure



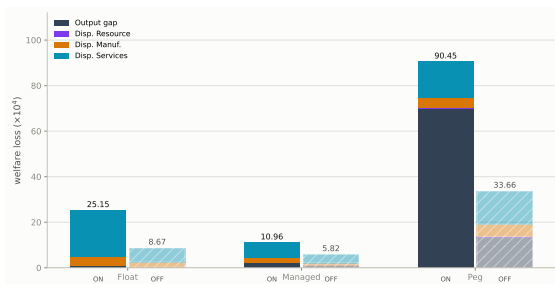
# Reading the Robustness Slides: What Varies, What's Fixed

Every comparison slide that follows dials **one** object and holds the rest of the calibration fixed — but "fixed" doesn't always mean "untouched": some dials force a mechanical rebalancing elsewhere so cost shares still sum to one. This is the map.

| Slide / dial                                                       | Varies                                                      | Held fixed / mechanically adjusted                                                                                                     |
|--------------------------------------------------------------------|-------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------|
| Isolating the Network Channel; $\phi_s$ vs. network; Why Services? | cross-sector density $\rho$ (off-diag. $\Omega^H$ )         | own-input $\Omega_{ii}^H$ , imports $\Omega_i^F$ fixed; $\alpha_i$ ( <b>labor share</b> ) <b>absorbs the change</b> so shares sum to 1 |
| Import / Export Openness                                           | $\Omega_i^F$ (or export share) scale                        | everything else fixed at Chile baseline; $\alpha_i$ <b>absorbs</b>                                                                     |
| Risk-Premium Volatility                                            | $sd(\varepsilon^{RP})$ only                                 | persistence $\rho_{RP}=0.80$ , every other shock, all cost shares — nothing else moves                                                 |
| Korea / Czechia                                                    | <b>entire calibration</b> ( $\Omega^H, \Omega^F, \beta^H$ ) | (all only functional-form parameters ( $\gamma, \varphi, \varepsilon, \beta$ ) shared with Chile                                       |

**Not yet swept:** the debt-elastic risk-premium *elasticity*  $\psi$  itself (only its shock's *volatility* is, above) — flagged as the one remaining diagnostic on the "75% risk-premium share" result.

# Welfare Ranking: Managed Float Dominates



**Managed < Float  $\ll$  Peg:**  $10.96/25.15/90.45$  ( $\times 10^{-4}$ )

**Float & Managed:** almost entirely price dispersion, concentrated in Services

**Peg:** almost entirely output gap

**Without the network:** same ranking, but the network roughly **doubles-to-triples** every regime's loss

**Units:**  $\mathcal{W} = \frac{1}{2}[(\gamma + \varphi)\text{Var}(\check{y}_t) + \sum_i \kappa_i \text{Var}(\pi_{it})]$  is a **per-period, consumption-equivalent fraction** (standard Rotemberg–Woodford quadratic-loss normalization) — Managed's  $10.96 \times 10^{-4}$  means an average per-period loss of  $\approx 0.11\%$  of steady-state consumption, every period, forever, from stabilization policy alone (on top of any level effects, which this second-order

# Counting the Cross-Sector Term (Rubbo Prop. 3)

Christian: *"I really need to count for that."* Rubbo's welfare formula has a **between-sector** piece,  $\Phi_C + \sum_s \lambda_s \Phi_s$  (relative-price misallocation across sectors), that every number so far omits. Now built and computed, not just flagged.

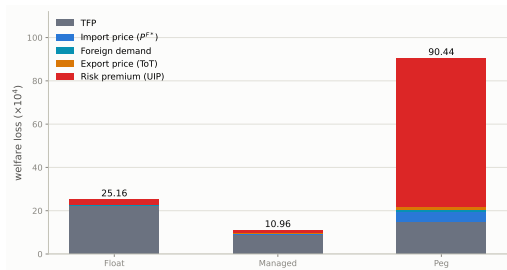
**Closed form:** both cross-sector aggregators here are **Cobb-Douglas** (household consumption across domestic sectors  $\beta^H$ ; firm input bundles  $\Omega^H, \Omega^F$ ) — Cobb-Douglas has Allen-Uzawa elasticity of substitution *exactly* 1 by construction, so  $\Phi_C, \Phi_s$  collapse to a **weighted variance of the markup gap**  $\mu_{it} = \log(P_{it}/MC_{it}) - \log \frac{\varepsilon}{\varepsilon-1}$  across sectors (only the outer  $C^H/C^F$  nest is genuine CES,  $\eta=1.5$ , used for the one domestic-vs-import pair).  $\mu_{it}$  added as a pure reporting definition to the .mod files — changes nothing else.

|                                                       | Float        | Managed      | Peg           |
|-------------------------------------------------------|--------------|--------------|---------------|
| Within-sector + output gap (headline)                 | 25.47        | 10.17        | 102.05        |
| Cross-sector $\Phi_C + \sum_s \lambda_s \Phi_s$ (new) | 0.83         | 0.86         | 4.57          |
| <b>New total</b>                                      | <b>26.31</b> | <b>11.03</b> | <b>106.62</b> |
| % added                                               | 3.3%         | 8.4%         | 4.5%          |

**Ranking unchanged:** Managed < Float ≪ Peg, same as before — the cross-sector term is real (not zero) but **second-order** relative to within-sector dispersion, because it's mechanically bounded by the *spread* across only 3 sectors' price gaps, not any single sector's own variance. **Caveat:**

domestic vs import cross elasticity uses  $\eta$  as a practical stand in for the true nested CES

# What Drives Each Regime's Loss? Shock Decomposition



**Peg:** risk-premium/UIP  $\rightarrow$  76% of loss, mostly output gap

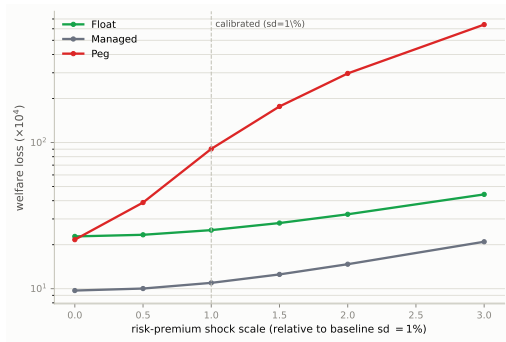
**Float & Managed:** TFP shocks dominate instead — risk-premium barely registers since the exchange rate (Float) or partial smoothing (Managed) absorbs it before it reaches output

Import price, foreign demand, and export price are second-order for every regime

**How to read this:** allocates *total welfare loss* (previous slide) across shocks using Dynare's own variance decomposition, exactly — %-of-variance-by-shock  $\times$  that variable's total variance, summed with the same  $\kappa_i/(\gamma+\varphi)$  weights as the welfare formula, so shock contributions add up

# How Much Does Peg's Loss Depend on the Risk-Premium Shock?

Adam: "effects of the volatility of the exchange-rate processes." Risk-premium/UIP drives 76% of Peg's loss (previous slide) — calibrated at the **same 1% s.d.** as every other shock, no independent target. **Important, uncomfortable finding, stated plainly:**

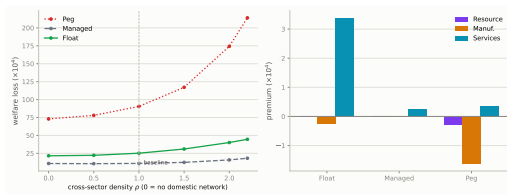


**With the risk-premium shock switched off entirely, Peg is *not* worst — Peg  $0.95 \times$  Float (essentially tied, Peg fractionally *better*). The entire "Peg is dominated" headline rests on this one shock existing, not on terms-of-trade, import pass-through, or any network channel**

Turn it on at **half** the calibrated size and Peg is already  $1.7 \times$  Float; at the **calibrated** size,

# Isolating the Network Channel

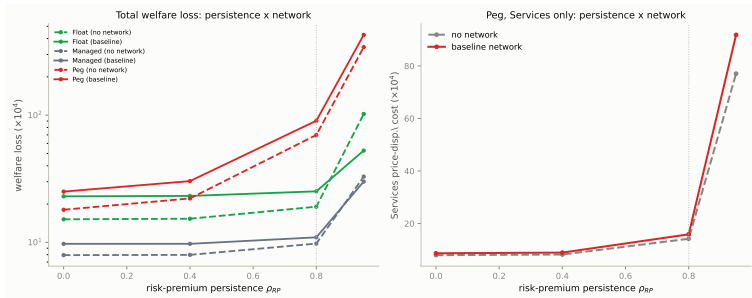
Does the network's effect scale *continuously* with how interconnected the economy is?



Denser network amplifies shock propagation for *everyone*, but **Peg** has no policy lever ( $\phi_\pi, \phi_y = 0$ ) to lean against it, so the amplification passes straight into welfare loss; **Float/Managed** actively damp part of it with the interest rate — **gap widens** as  $\rho \uparrow$

# Persistence, Not Just Size: Why UIP Losses Are So Large

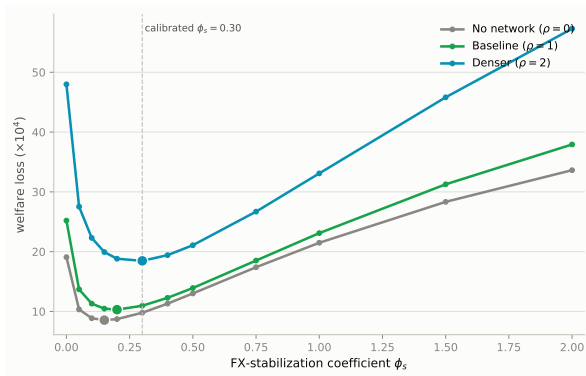
Christian: "*persistence of inflation leading to higher cost.*" The risk-premium/UIP shock's persistence  $\rho_{RP}$  (baseline 0.80) — crossed with network density — turns out to matter **far more convexly** than its size (previous risk-premium slide) or the network alone.



AR(1) variance scales as  $\sigma^2/(1 - \rho_{RP}^2)$  — **mechanically convex**, not linear, in persistence. Moving  $\rho_{RP}$  from 0.0  $\rightarrow$  0.40  $\rightarrow$  0.80 raises Peg's loss only modestly ( $18 \rightarrow 22 \rightarrow 70 \times 10^{-4}$  at  $\rho=0$ ), but 0.80  $\rightarrow$  0.95 **explodes it** to  $343 \times 10^{-4}$  — a 5x jump from a 0.15 change in persistence, not size

# Does the Optimal Managed-Float Response Depend on the Network?

Christian's question: welfare loss changes a lot with network density — does the **welfare-minimizing** FX-stabilization coefficient  $\phi_s$  change too, or just its magnitude?



Re-solves the managed-float rule  $\hat{t}_t = \phi_\pi \pi_t^{DC} + \phi_y \tilde{y}_t + \phi_s \log S_t$  over the  $\phi_s$  grid at three network densities: **no network** ( $\rho=0$ ), **baseline** ( $\rho=1$ ), **denser** ( $\rho=2$ )

$\phi_s^*$  **rises with density**: 0.15  $\rightarrow$  0.20  $\rightarrow$  0.30 — denser network, more indirect cost-push (previous slide), so the welfare-minimizing rule leans harder against the exchange rate

# How the Network Amplifies Shocks

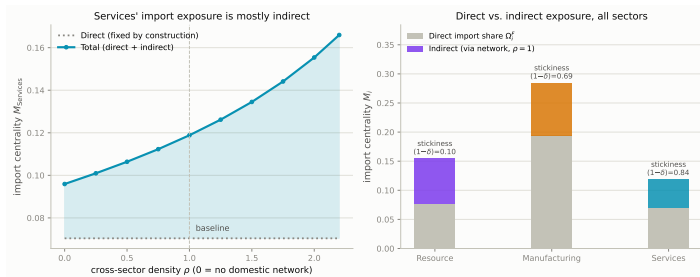
The network **dampens** the inflation peak but **amplifies** the output-gap peak — in every regime, for *both* shocks.

| Shock        | Regime  | Inflation peak<br>(Network ON ÷ Network OFF) | Output-gap peak |
|--------------|---------|----------------------------------------------|-----------------|
| Import-price | Float   | 0.40×                                        | 2.77×           |
|              | Managed | 0.61×                                        | 2.28×           |
|              | Peg     | 0.86×                                        | 3.15×           |
| Risk-premium | Float   | 0.41×                                        | 2.57×           |
|              | Managed | 0.57×                                        | 0.98×           |
|              | Peg     | 0.67×                                        | 2.16×           |

**Note:**  $< 1\times$  = network shrinks the peak;  $> 1\times$  = network amplifies it. (Peak IRF response over 20 quarters; exchange-rate peak itself barely moves with the network)

# Why Services? Direct vs. Indirect Import Exposure

Services is the **stickiest** sector (Calvo reset prob.  $\delta_3=0.16$ ) and has the **lowest direct** import share ( $\Omega_3^F=0.070$ ) — yet it buys 5.8% of its cost from Manufacturing, the most import-intensive sector ( $\Omega_2^F=0.195$ ).



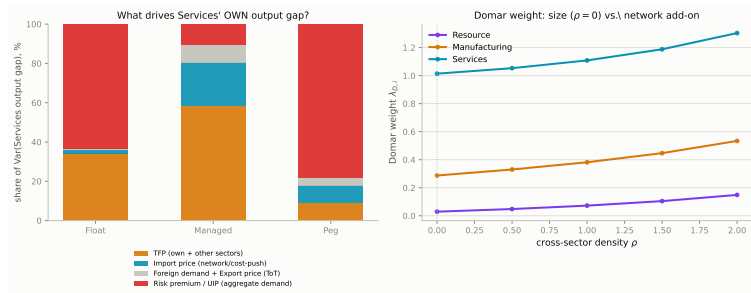
Total import centrality  $M_i = [(I - \Omega^H)^{-1} \Omega^F \mathbf{1}]_i$  splits into a **direct** part (own  $\Omega_i^F$ ) and an **indirect** part inherited from suppliers — for Services, **41% of total exposure is indirect** (Resource 50%, Manuf. 32%)

Indirect exposure is **mechanically zero** at  $\rho=0$  and grows with network density  $\rho$  — density doesn't change any sector's *direct* import share, only how much of *other* sectors' import content gets passed to it

**Combined with stickiness:** Services can't reprice quickly, so whatever cost-push it inherits shows up as persistent inflation, not a one-quarter blip —  $\text{Corr}(\pi_3, \pi_3^{t-1}) \approx 0.82$ , the most persistent inflation series in the model

# Which Channel Actually Drives Services' Exposure?

Is it the import network, monetary mismatch, or just size? Dynare's own shock-by-shock variance decomposition of Services' **own** output gap says: mostly **neither** of the first two alone.



**Left:** the import-price/network channel (previous slide) is real but **secondary** for Services' output gap — 2% (Float) / 22% (Managed) / 9% (Peg) of its variance. **Risk-premium/UIP** — a pure aggregate, non-import shock — dominates instead: 63% (Float), 78% (Peg)

**Right:** Services' Domar weight  $\lambda_{D,3}$  (its welfare-relevant "size") is 92% already present at  $\rho=0$  — from its own consumption share  $\beta_3^H=0.74$  plus heavy *within*-sector input reuse ( $\Omega_{33}^H=0.27$ ) — the cross-sector network adds only  $\approx 9\%$  more. Contrast Resource: network more than doubles its weight (60% of its total is network-added)

**Synthesis** — all three of Christian's hypotheses compound, none is "the" answer: size

# Network Exposure & Sector Welfare Preferences

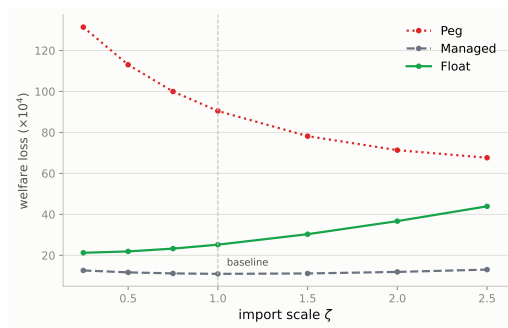
Each cell: **Network ON** / **Network OFF**.

| Price-dispersion cost ( $\times 10^4$ ) |              |             |               |
|-----------------------------------------|--------------|-------------|---------------|
| Sector                                  | Float        | Managed     | Peg           |
| Resource                                | 0.05 / 0.03  | 0.04 / 0.02 | 0.27 / 0.27   |
| Manuf.                                  | 3.87 / 2.23  | 1.92 / 0.94 | 4.28 / 5.32   |
| Services                                | 20.25 / 6.22 | 6.71 / 3.74 | 15.88 / 14.59 |

| Var(own output gap) ( $\times 10^4$ ) |             |             |              |
|---------------------------------------|-------------|-------------|--------------|
| Sector                                | Float       | Managed     | Peg          |
| Resource                              | 0.13 / 0.02 | 0.14 / 0.04 | 0.20 / 0.12  |
| Manuf.                                | 0.58 / 0.16 | 0.38 / 0.08 | 1.98 / 0.23  |
| Services                              | 0.38 / 0.15 | 2.31 / 1.33 | 35.16 / 8.97 |

**Every sector agrees:  $\text{Managed} < \text{Float} < \text{Peg}$**  — Services' huge stake in avoiding Peg is mostly **network-driven** (output-gap cost  $8.97 \rightarrow 35.16$  with the network on).

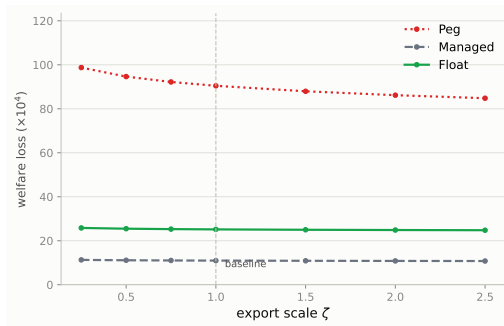
# Import Openness



## Managed < Float < Peg robust throughout

More import exposure means more  $\Gamma \Delta e_t$  cost-push for **Float** (loss rises), but that channel is structurally zero under **Peg** ( $\hat{s}_t \equiv 0$ ), so its loss instead falls via the labor-share renormalization — **gap converges**

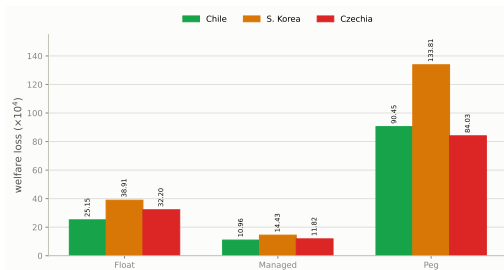
# Export Openness



**Managed < Float < Peg robust throughout**

Exports work only through **demand** ( $EX_{it} \rightarrow \tilde{y}_t$ ), never marginal cost — a pure buffer, so **all regimes improve**, and **Peg gains most** since it has no FX buffer of its own

# Robustness: South Korea & Czechia



**Managed < Float  $\ll$  Peg in every calibration**

Korea higher throughout: denser network & higher import exposure

# Full Native Resolution: Same Ranking, Bigger Margin

Re-solved at each country's full IO resolution ( $N = 12$  Chile, 33 Korea, 81 Czechia); welfare loss  $\times 10^{-4}$ :

|         |          | Float | Managed | Peg   | Peg/Float |
|---------|----------|-------|---------|-------|-----------|
| Chile   | $N = 3$  | 25.5  | 10.2    | 102.1 | 4.0×      |
|         | $N = 12$ | 15.9  | 9.4     | 108.3 | 6.8×      |
| Korea   | $N = 3$  | 38.9  | 14.4    | 133.8 | 3.4×      |
|         | $N = 33$ | 13.0  | 9.4     | 133.3 | 10.3×     |
| Czechia | $N = 3$  | 32.2  | 11.8    | 84.0  | 2.6×      |
|         | $N = 81$ | 11.7  | 8.2     | 82.1  | 7.0×      |

**Managed** < **Float**  $\ll$  **Peg** survives in all three countries — no reversal

**Peg's loss is unchanged** (+6%, -0.4%, -2%); **Float falls 37–67%**, Managed 8–35%

Risk premium still drives 78–83% of Peg's output-gap variance; TFP (all  $N$  sectors) < 2.2%

# Why the Margin Grows: Diversification of Sectoral Shocks

**Risk premium = one aggregate shock:** enters via UIP once, whatever  $N$  is  $\Rightarrow$  Peg's loss is invariant to resolution

**Float/Managed loss is 81–92% sectoral TFP** (price-dispersion term). With every sector given the same 1% s.d.,  $N$  small independent shocks replace 3 large correlated ones:  $\sum_i \lambda_{D,i}^2$  falls to 28% / 11% / 14% (Chile / Korea / Czechia) — Gabaix (2011)

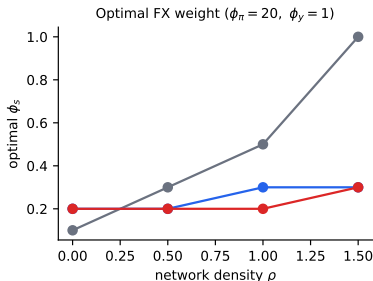
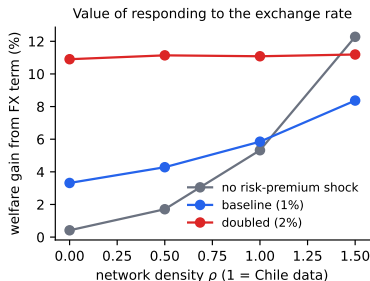
**Test:** rescale sectoral TFP s.d. to restore the 3-sector aggregate TFP variance

|                          | Float | Managed | Peg output-gap term |
|--------------------------|-------|---------|---------------------|
| Korea, baseline $N = 33$ | 13.0  | 9.4     | 113.7               |
| Korea, TFP rescaled      | 88.9  | 65.7    | 121.4               |

Float/Managed rise several-fold; Peg's risk-premium-driven term moves  $\leq 7\%$  (over-corrects: an upper bound)

**Caveat:** the *level* of Float/Managed loss depends on how sectoral shock volatility is normalized — not disciplined by IO data

# Optimal FX Weight: With vs. Without the Network



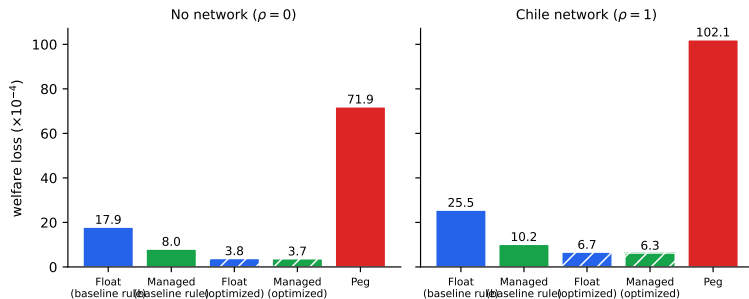
Joint optimum over  $(\phi_\pi, \phi_y, \phi_s)$ :  $\phi_\pi \approx 20$ ,  $\phi_y \approx 1$  — near strict DC-index targeting;  $\phi_s$  is a small correction

**Real shocks:** FX term worth 0.4% (no network)  $\rightarrow$  12% ( $\rho = 1.5$ );  $\phi_s^*$ : 0.1  $\rightarrow$  1.0

**Risk-premium shock:** FX term worth  $\approx 11\%$  (doubled shock) at every  $\rho$ ,  $\phi_s^* \approx 0.2$

**Baseline shocks:** gain only 3–8%; loss is flat in  $\phi_s$  (0.2–0.3 within 1%)

# Regimes at Optimized Rules: What Survives



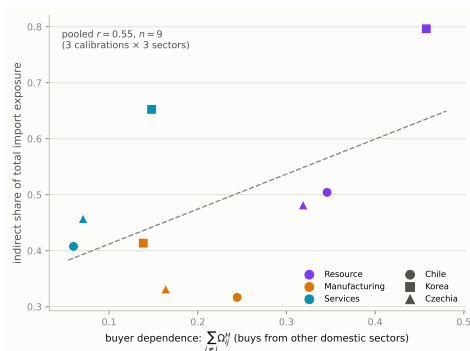
“Managed beats Float 2.5 $\times$ ” compares a *weak* float rule ( $\phi_\pi = 1.5$ ) with an FX term that substitutes for it

Both optimized: Managed beats Float by only **3% (no network)** – **6% (Chile)**

**Peg is 16–20 $\times$  worse than an optimized rule**, with or without the network (still 4 $\times$  with no risk-premium shock)

# Robustness: Is Indirect Exposure a General Pattern?

Adam's question: is "low direct import, high indirect import" (Services, two slides back) a general downstream-sector pattern, or a one-off? Pools **Chile, Korea, Czechia** × 3 sectors — 3 independent national IO tables, not the same data reused.



**Honest answer: only a moderate, noisy relationship** ( $r=0.55$ ,  $n=9$ ) between how much a sector sources domestically and how much of its import exposure is indirect — **with just 3 sectors this is a weak test**, not a proven general law

**What IS robust across all three calibrations:** Services sources the *least* domestically of any sector in every single calibration (lowest buyer-dependence, Chile/Korea/Czechia), yet still derives 41–65% of its import exposure indirectly — because what little it buys is

# Conclusion

**The network matters for severity and incidence, not the ranking:** switching it on raises welfare losses in every regime and reallocates who bears them, but **Managed < Float < Peg either way**

**Managed float wins** because it dampens the FX pass-through channel without fully reimposing peg's loss of autonomy

**Robust** across three real-data calibrations (Chile, Korea, Czechia) and a genuine second-order solution

**Optimal rule:** near strict DC-index targeting; the FX term is worth 3–8% at baseline, more with a dense network (real shocks) or large UIP shocks; managed vs. float shrinks to 3–6% when both are optimized, peg stays far worse

**Full native resolution** ( $N = 12, 33, 81$ ): ranking unchanged; Peg's loss invariant to resolution, Float's falls 37–67% (diversification of sectoral shocks) — so the Peg/Float *margin* is not pinned down without an estimated risk-premium process